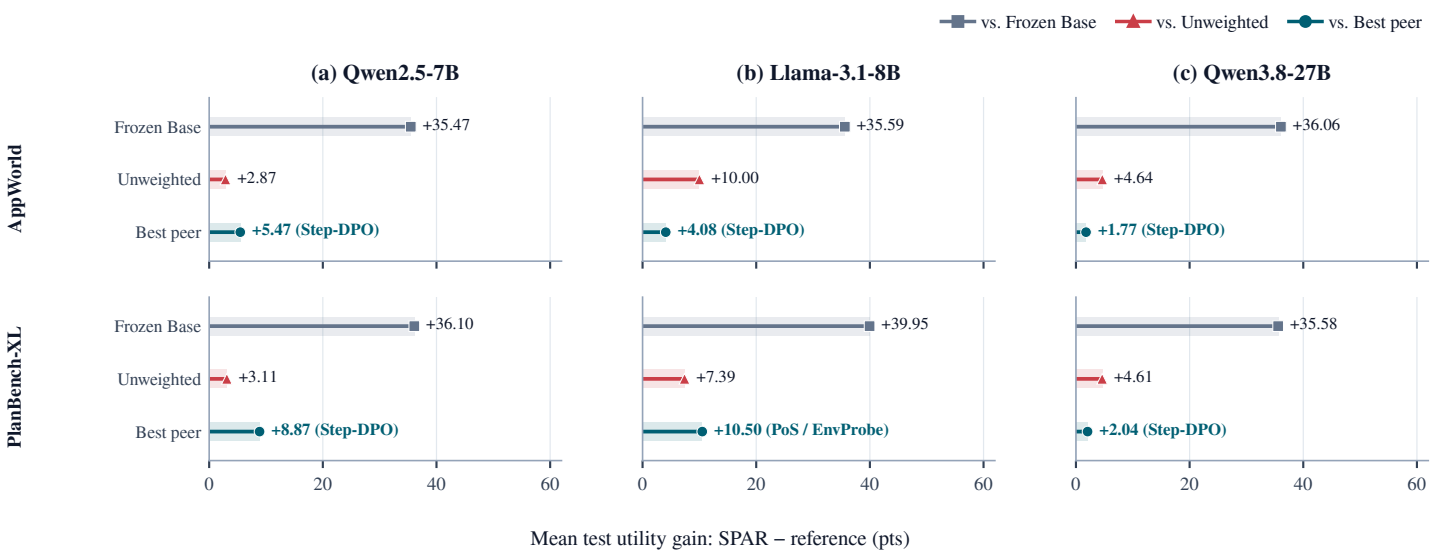
vs. Frozen Base
vs. Unweighted
vs. Best peer
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
AppWorld
PlanBench-XL
Frozen Base
Unweighted
Best peer
+35.47
+2.87
+5.47 (Step-DPO)
+35.59
+10.00
+4.08 (Step-DPO)
+36.06
+4.64
+1.77 (Step-DPO)
+36.10
+3.11
+8.87 (Step-DPO)
+39.95
+7.39
+10.50 (PoS / EnvProbe)
+35.58
+4.61
+2.04 (Step-DPO)
0
20
40
60
Mean test utility gain: SPAR − reference (pts)